

Topic:

Hybrid Renewable Energy Sources

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Abstract:

Hybrid Renewable Energy Systems (HRES) combine two or more renewable energy sources such as solar, wind, biomass, and small hydro to generate reliable and efficient electricity. These systems are designed to overcome the limitations of individual renewable sources, especially their intermittent nature. For example, solar energy is available only during daytime, while wind energy may be stronger at night or during certain seasons. By integrating multiple sources along with energy storage systems like batteries, hybrid systems ensure continuous power supply, improved efficiency, and better grid stability.

Hybrid systems are widely used in rural electrification, remote areas, and smart grid applications. They reduce dependence on fossil fuels, lower greenhouse gas emissions, and promote sustainable development. Advanced technologies such as power electronics and energy management systems help optimize power generation and distribution. Although the initial investment cost may be higher, hybrid renewable systems provide long-term economic and environmental benefits. Therefore, Hybrid Renewable Energy Sources offer a reliable, eco-friendly, and sustainable solution to meet the increasing global energy demand.